

Collections

Name: O. Bhanu Prakash Reddy

Course Code: CSA0904

Reg No: 192372075

Course name: programming In Java

1.A company HR department wants to store the names of employees who attended a training program.

Question

Write a Java program to:

Create an ArrayList<String>

Add employee names: Ramesh, Suresh, Priya, Anu, Karthik

Display all employee names in the order they were added.

Code:

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        ArrayList<String> employees = new ArrayList<>();
```

```
        System.out.println("Enter the names of 5 employees:");
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            System.out.print("Employee " + i + ": ");
```

```
            employees.add(sc.nextLine());
```

```
}
```

```
System.out.println("\nEmployees who attended the training program:");
```

```
for (String employee : employees) {
```

```
    System.out.println(employee);
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

Input:

```
Enter the names of 5 employees:
Employee 1: Ramesh
Employee 2: Suresh
Employee 3: Priya
Employee 4: Anu
Employee 5: karthik
```

Output:

```
Employees who attended the training program:
Ramesh
Suresh
Priya
Anu
karthik
```

2. Restaurant Order Management (ArrayList)

Scenario

A restaurant stores the food items ordered by a customer.

Question

Write a Java program to:

Create an ArrayList<String>

Add items: Dosa, Idli, Pongal, Coffee

Remove Pongal

Display the updated order list and the total number of items.

Code:

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        ArrayList<String> orderList = new ArrayList<>();
```

```
        System.out.println("Enter 4 food items:");
```

```
        for (int i = 1; i <= 4; i++) {
```

```
            System.out.print("Item " + i + ": ");
```

```
            orderList.add(sc.nextLine());
```

```
        }
```

```
        System.out.print("\nEnter the food item to remove: ");
```

```
        String item = sc.nextLine();
```

```
        if (orderList.remove(item)) {  
            System.out.println(item + " removed successfully.");  
        } else {  
            System.out.println(item + " not found.");  
        }  
  
        System.out.println("\nUpdated Order List:");  
  
        for (String itemName : orderList) {  
            System.out.println(itemName);  
        }  
  
        System.out.println("Total Number of Items: " + orderList.size());  
  
        sc.close();  
    }  
}
```

Input:

```
Enter 4 food items:  
Item 1: Dosa  
Item 2: Idli  
Item 3: Pongal  
Item 4: Coffee
```

Output:

```
Enter the food item to remove: Pongal  
Pongal removed successfully.  
  
Updated Order List:  
Dosa  
Idli  
Coffe  
Total Number of Items: 3
```

3. Online Course Enrollment (ArrayList)

Scenario

An online learning platform stores the names of students enrolled in a Java course.

Question

Write a Java program to:

Store 5 student names in an ArrayList

Check whether Meena is enrolled

Print Enrolled if the name exists; otherwise print Not Enrolled.

Code:

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        ArrayList<String> students = new ArrayList<>();
```

```
        System.out.println("Enter the names of 5 students:");
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            System.out.print("Student " + i + ": ");
```

```
            students.add(sc.nextLine());
```

```
        }
```

```
        System.out.print("\nEnter the student name to search: ");
```

```
        String name = sc.nextLine();
```

```
        if (students.contains(name)) {  
            System.out.println("Enrolled");  
        } else {  
            System.out.println("Not Enrolled");  
        }  
  
        sc.close();  
    }  
}
```

Input:

```
Enter the names of 5 students:  
Student 1: Suresh  
Student 2: ramesh  
Student 3: meena  
Student 4: kishore  
Student 5: king
```

Output:

```
Enter the student name to search: meena  
Enrolled
```

4. Unique Email Registration (Set)

Scenario

A website should not allow duplicate email registrations.

Question

Write a Java program to:

Create a HashSet<String>

Add emails: a@gmail.com, b@gmail.com, a@gmail.com, c@gmail.com

Display all unique email addresses

Display the total number of unique registered users.

Code:

```
import java.util.HashSet;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        HashSet<String> emailSet = new HashSet<>();
```

```
        System.out.print("Enter the number of email addresses: ");
```

```
        int n = sc.nextInt();
```

```
        sc.nextLine();
```

```
        System.out.println("Enter the email addresses:");
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print("Email " + i + ": ");
```

```
            emailSet.add(sc.nextLine());
```

```

    }

    System.out.println("\nUnique Email Addresses:");

    for (String email : emailSet) {
        System.out.println(email);
    }

    System.out.println("Total Number of Unique Registered Users: " + emailSet.size());

    sc.close();
}
}

```

Input:

```

Enter the number of email addresses: 4
Enter the email addresses:
Email 1: a@gmail.com
Email 2: b@gmail.com
Email 3: a@gmail.com
Email 4: c@gmail.com

```

Output:

```

Unique Email Addresses:
b@gmail.com
c@gmail.com
a@gmail.com
Total Number of Unique Registered Users: 3

```


5. Library Book ID Tracking (Set)

Scenario

A library stores book ID numbers. The same book ID should not be entered twice.

Question

Write a Java program to:

Create a `HashSet<Integer>`

Add book IDs: 101, 102, 103, 101, 104

Display all unique book IDs

Print Duplicate Book ID Found when a duplicate ID is added.

Code:

```
import java.util.HashSet;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        HashSet<Integer> bookIds = new HashSet<>();
```

```
        System.out.print("Enter the number of Book IDs: ");
```

```
        int n = sc.nextInt();
```

```
        System.out.println("Enter the Book IDs:");
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print("Book ID " + i + ": ");
```

```
        int id = sc.nextInt();

        if (bookIds.add(id)) {
            System.out.println("Book ID Added");
        } else {
            System.out.println("Duplicate Book ID Found");
        }
    }

    System.out.println("\nUnique Book IDs:");

    for (int id : bookIds) {
        System.out.println(id);
    }

    sc.close();
}
}
```

Input:

```
Enter the number of Book IDs: 5
Enter the Book IDs:
Book ID 1: 101
Book ID Added
Book ID 2: 102
Book ID Added
Book ID 3: 101
Duplicate Book ID Found
Book ID 4: 103
Book ID Added
Book ID 5: 104
Book ID Added
```

Output:

```
Unique Book IDs:
101
102
103
104
```

